



• AI-POWERED DETECTION

Real-Time Accident Detection Using Computer Vision

Leveraging YOLOv8 deep learning to instantly detect car accidents in video feeds or live streams – enabling faster emergency response and saving lives.

Login

How It Works

PROCESS

How It Works

From input to detection in milliseconds – here's the pipeline behind our accident detection system.

01

Input Source

Upload a video file (dashcam, surveillance, live streams, or any recorded footage).

02

Frame Extraction

The video is split into individual frames, processed every 2nd frame for efficiency.

03

YOLO Inference

Each frame passes through the YOLOv8 model, which predicts bounding boxes and classes.

04

Alert & Display

Accidents are highlighted with red boxes, warning banners, and instant visual alerts.

FEATURES

Intelligent Accident Detection

Our system combines cutting-edge computer vision with deep learning to provide accurate, real-time accident detection.

AI

YOLOv8 Detection

Powered by the YOLOv8 architecture – one of the most advanced real-time object detection models. Trained specifically on accident scenarios for maximum accuracy.

UP

Video Upload Analysis

Upload surveillance videos, live streams, or any recorded video file. The system processes each frame and highlights detected accidents with bounding boxes.

AL

Audio Warning Alerts

When an accident is detected, the system plays an audio warning sound in the browser along with visual alerts – ensuring no incident goes unnoticed.

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AL**Audio Warning Alerts**

When an accident is detected, the system plays an audio warning sound in the browser along with visual alerts – ensuring no incident goes unnoticed.

IN**Instant Alerts**

The moment an accident is detected, a visual warning banner appears with a flashing red border – ensuring the operator is immediately notified.

MC**Multi-Class Detection**

The model distinguishes between normal cars and accident scenes, drawing green boxes for vehicles and red boxes with high-visibility labels for detected accidents.

WB**Web-Based Interface**

No software installation required. Access the detection system from any browser – upload videos or use the live stream feature and get instant results through an intuitive web dashboard.



Choose Your Role

This choice decides which workspace you will use after registration and sign in.



Admin / Operator

Run detection, review incidents, dispatch alerts, and monitor response status.



Responder

Receive assigned alerts, update response progress, and resolve active cases.

Alert Management

Monitoring Active



Emergency Mode Active

Review new incidents, report them to authorities, and monitor responder progress.

Broadcast Emergency Protocol



ACTIVE

73A2D0EB

Airport Road Camera 04

Incident reported to responders. Current responder status: Pending.

29 seconds ago
DETECTED TIME

Detected: 2026-05-20 01:35:54

Assigned: Responder Team

Responder: Pending

Status: Active

Awaiting Authority Contact

Sent to Responder

False Alarm



ACTIVE

813AC3CD

Airport Road Camera 04

Incident reported to responders. Current responder status: Pending.

1 day ago
DETECTED TIME

Detected: 2026-05-18 22:56:05

Assigned: Responder Team

Responder: Pending

Status: Active

Awaiting Responder

Sent to Responder

False Alarm

ACCIDENT DETECTED - RESPONDER

Dashboard

System Online 01:37:49 AM

ACTIVE ALERTS

4

Open incidents not yet closed

CAMERAS ONLINE

1/165

Active and reachable devices

EVENTS TODAY

1

Recorded incidents in the last 24 hours

AVG MODEL DETECTION TIME

282ms

Average inference time per saved detection

Recent Events

View All

EVENT ID	LOCATION	SEVERITY	STATUS	TIME
EVT-20260520-001	Monitored Zone 1	CRITICAL	Pending	1 minute ago
EVT-20260518-002	Monitored Zone 2	CRITICAL	Pending	1 day ago
EVT-20260517-003	Monitored Zone 3	CRITICAL	Pending	2 days ago
EVT-20260508-004	Monitored Zone 4	CRITICAL	Pending	12 days ago
EVT-20260506-005	Monitored Zone 1	LOW	False Alarm	13 days ago
EVT-20260506-006	Monitored Zone 2	LOW	False Alarm	13 days ago

Responder Dashboard

Ready for Dispatch

ACTIVE ALERTS

3

^

Open incidents not yet closed



TODAY'S CASES

1

^

Recorded incidents in the last 24 hours



ASSIGNED ALERTS

50

All alerts assigned to responders



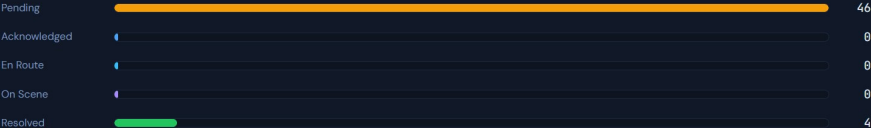
PENDING ALERTS

46

Waiting for first response



Alerts by Response Status



Assigned Alerts

ALERT ID	LOCATION	DETECTED	ASSIGNED TO	STATUS	ACTION
73A200EB	Airport Road Camera 04	9 minutes ago	mahabntfaisal6@gmail.com	RESOLVED	View / Update

Responder Alerts

Assigned Cases

Review each responder case, open details, and update the response status.

RESOLVED

73A2DDEB

STATUS

Closed

DETECTED

2026-05-20 01:35:54

LOCATION

Airport Road Camera 04



RESOLVED

813AC3CD

STATUS

Closed

DETECTED

2026-05-18 22:56:05

LOCATION

Airport Road Camera 04



RESOLVED

F3F6380C

STATUS

Closed

DETECTED

2026-05-17 21:11:51

LOCATION

Downtown Sector 2



RESOLVED

0916FA93

STATUS

Closed

DETECTED

2026-05-08 01:15:47

LOCATION

King Fahd Road - Northbound



RESOLVED

ABF5060C

STATUS

Closed

DETECTED

2026-05-04 11:34:14

LOCATION

Main St & 4th Ave



RESOLVED

C5205D35

STATUS

Closed

DETECTED

2026-05-04 11:30:51

LOCATION

Airport Road Camera 04



RESOLVED

87F3AC33

STATUS

Closed



PENDING

8B9798CF

STATUS

Closed



PENDING

B8A18D8E

STATUS

Closed



Responder Alerts

Accident Image or Video Preview

Enlarge Image



Full Alert Information

CURRENT STATUS

Resolved

LOCATION

Airport Road Camera 04

DETECTED TIME

2026-05-20 01:35:54 (11 minutes ago)

ALERT ID

73a2ddeb

ASSIGNED EMAIL

m*****3@gmail.com

Update Response

Acknowledge

En Route

On Scene

Resolved